

Cracking Query Bottlenecks: Towards Efficiency-Oriented Text-to-SQL Generation

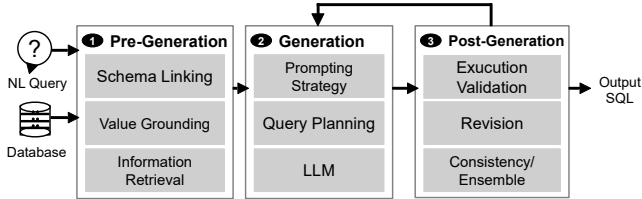


Fig. 1: Overview of a typical Text-to-SQL pipeline, illustrating the Pre-Generation, Generation, and Post-Generation stages

Abstract—Pending...

I. INTRODUCTION

Text-to-SQL aims to translate natural language (NL) questions into executable SQL queries, enabling non-expert users to retrieve and analyze relational data without manually writing database programs [1], [2], [3]. Its importance has grown with increasingly large schemas, rich database contents, and diverse analytical tasks, motivating benchmarks such as SPIDER [4], BIRD [5]. Recent LLM-based systems have achieved substantial progress through in-context learning, task decomposition and specialized agents, moving the field beyond single-shot generation toward workflow-based SQL construction [6], [7], [8], [9].

From Single-shot Generation to Agentic Workflows. As shown in Figure 1, modern Text-to-SQL systems typically organize SQL construction into three stages: pre-generation, generation, and post-generation. In the pre-generation stage, the system grounds the natural language query in the database context through schema linking, value grounding, and information retrieval, thereby reducing the search space and enriching the input evidence [9], [10], [11], [12]. In the generation stage, the model applies prompting strategies, query planning, and LLM-based reasoning to produce one or more candidate SQL queries [6], [7], [8], [13], [14], [15]. In the post-generation stage, these candidates are further checked and refined through execution validation, repair, and consistency- or ensemble-based selection [16], [7], [17], [8], [9], [13], [14], [15]. Compared with single-pass generation, such workflows are more robust because they decompose SQL construction into modular steps, strengthen database grounding, diversify candidate generation, and provide post-hoc opportunities for validation and refinement. Despite these advances, existing Text-to-SQL systems still face three coupled limitations:

L1: Limited Adaptation in Workflows and Strategy Selection. Many existing Text-to-SQL agents often predefine

both the workflow structure and the strategy choices, where modules such as schema linking, few-shot retrieval, question decomposition, candidate generation, validation, and revision are applied in a largely fixed manner [6], [7], [8], [13], [11]. Although such designs improve robustness over single-shot generation, they provide limited support for adapting reasoning strategies to the current database environment, question difficulty, or intermediate feedback. This can misallocate reasoning effort: simple queries over intuitive schemas may incur unnecessary token and latency overhead, while difficult queries may require deeper grounding, candidate comparison, or additional validation and reflection. This mismatch is illustrated by the different effects of domain-specific hints in BIRD [5]. For an intuitive database such as *superhero*, where tables and columns are relatively descriptive, adding domain hints provides little additional benefit in our experiments. In contrast, for a domain-heavy database such as *thrombosis_prediction*, where terms such as LDH and gender-dependent uric-acid ranges require medical knowledge, the same hints improve accuracy from 20% to nearly 50%. This contrast suggests that Text-to-SQL agents should adapt their grounding and generation strategies to the database environment, rather than applying the same strategy uniformly.

L2: Post-hoc rather than Process-level Reflection. Feedback in many Text-to-SQL systems is mainly used after a candidate SQL has been generated, such as repairing execution errors, revising invalid queries, or selecting among completed candidates [7], [8], [9], [18]. Such feedback improves final-output validation, but provides limited support for monitoring and regulating the generation process itself. In particular, the agent often lacks a controller that continuously tracks whether schema linking is reliable, whether candidate disagreement is increasing, whether repeated repairs are converging, or whether the current strategy should be changed. As a result, early mistakes in schema linking, value grounding, or query planning may propagate into later steps and become cascading hallucinations. Moreover, without global process supervision, local repair actions may fix surface-level execution errors while leaving the overall reasoning trajectory insufficiently grounded in the user question and database context.

L3: Limited Awareness of Query Efficiency. Most existing Text-to-SQL systems are primarily optimized for correctness: they aim to generate SQL queries whose execution results match the ground-truth answers [19], [20], [21], [22], [23], [24], [6], [7]. However, in practical database applications, a semantically correct SQL query is not necessarily efficient. Equivalent or near-equivalent SQL formulations may lead to

very different execution costs due to redundant projections, unnecessary joins, correlated subqueries, missing filter push-down, suboptimal join orders, or functions applied to indexed columns [25], [26], [27]. Although database systems have long relied on rule-based, cost-based, and learned rewriting to improve query efficiency [28], [27], [29], [30], current Text-to-SQL agents usually use database feedback mainly to check executability or result consistency. They rarely treat execution plans, estimated costs, or bottleneck patterns as signals for deciding whether and how to rewrite the generated SQL. As a result, the final query may be correct but unnecessarily expensive, especially in enterprise settings with large schemas, large tables, and long multi-step analytical workflows [18].

Key Insights: Meta-cognitive and Environment-aware Text-to-SQL. Inspired by enactive cognition and meta-cognition, we view Text-to-SQL not as a static translation from question and schema to SQL, but as an adaptive interaction process between the agent and the database environment [31], [32]. Enactive cognition suggests that understanding is shaped through interaction with the environment, while meta-cognition emphasizes the ability to monitor and regulate one’s own reasoning process. In the context of Text-to-SQL, this means that an agent should perceive database-specific signals, act by generating and rewriting SQL, observe feedback from execution and query plans, and adjust its strategy when uncertainty, failure, or inefficiency is detected. These perspectives lead to three design objectives. First, to address **L1**, the agent should perform *environment-aware workflow adaptation via strategy selection*: it should select lightweight or deep workflows according to the perceived database environment, question complexity, and intermediate feedback, rather than applying the same strategy uniformly. Second, to address **L2**, the agent should support *process-level self-reflection and dynamic control*: it should monitor schema-linking reliability, candidate disagreement, repair convergence, and strategy suitability during generation, so that early mistakes can be corrected before they propagate. Third, to address **L3**, the agent should enable *execution- and plan-aware SQL optimization*: it should use query plans, cost signals, semantic validation, and reusable rewrite knowledge to improve query efficiency while preserving correctness.

II. MOTIVATION

A. Task Formulation

Traditional Text-to-SQL aims to translate a NL question x into an executable SQL query q over a database environment $\mathcal{D}$. Given a ground-truth query q^* , the standard objective is to generate a query whose execution result matches that of q^* :

$$f_\theta(x, \mathcal{D}) \rightarrow q, \quad \text{s.t.} \quad \text{Exec}(q, \mathcal{D}) = \text{Exec}(q^*, \mathcal{D}),$$

where f_θ denotes a Text-to-SQL model parameterized by θ , $\mathcal{D}$ denotes the database environment including schema, data, indexes, and statistics, and $\text{Exec}(q, \mathcal{D})$ denotes the result of executing q on $\mathcal{D}$.

However, this formulation focuses mainly on semantic correctness and does not explicitly consider query efficiency.

In practice, multiple semantically equivalent SQL queries may produce the same result but incur substantially different execution costs due to join order, predicate placement, aggregation strategy, or index utilization. We therefore formulate **Efficiency-Oriented Text-to-SQL**, where the goal is to generate a query that is both semantically correct and efficient. Given a NL question x and a database environment $\mathcal{D}$, the goal is to generate a SQL query q_{opt} satisfying:

$$q_{\text{opt}} = \arg \min_{q \in \mathcal{Q}_{\text{corr}}(x, \mathcal{D})} \text{Cost}(q, \mathcal{D}),$$

$$\mathcal{Q}_{\text{corr}}(x, \mathcal{D}) = \{q \in \mathcal{Q}(x, \mathcal{D}) \mid \text{Exec}(q, \mathcal{D}) = \text{Exec}(q^*, \mathcal{D})\},$$

where $\mathcal{Q}_{\text{corr}}(x, \mathcal{D})$ denotes the subset of candidates that are semantically correct with respect to q^* . $\text{Cost}(q, \mathcal{S})$ represents the estimated or actual execution cost of query q on $\mathcal{S}$. Intuitively, the model should not only ensure that q_{opt} yields correct results but also that it achieves minimal execution time or resource consumption under the same semantic constraints.

B. Design Objective

Existing Text-to-SQL systems are typically designed as static pipelines that decompose query generation into pre-defined stages, including schema linking, candidate generation, and post-hoc execution validation. While such designs improve robustness compared to single-pass generation, they lack the ability to dynamically adapt their reasoning process to different database environments, query difficulties, and intermediate execution feedback.

Beyond Static Pipelines: A Meta-Cognitive Multi-Agent Perspective. Inspired by enactive cognition and meta-cognitive theory, we reformulate Text-to-SQL as a dynamic interaction process between an intelligent system and the database environment. Enactive cognition emphasizes that intelligence emerges through continuous interaction with the environment, while meta-cognition enables the system to monitor, regulate, and adapt its own reasoning process. Based on these principles, we propose a Meta-Cognitive Multi-Agent Text-to-SQL System, where a central Meta-Cognitive Controller dynamically orchestrates a set of specialized agents (actions) under different execution contexts. Instead of a fixed pipeline, all stages in Text-to-SQL are decomposed into reusable and composable actions, which are invoked online based on the current state of the problem and environment feedback. This design establishes a unified closed-loop system in which a Meta-Cognitive Controller coordinates planning, execution, evaluation, and optimization over SQL programs based on continuous database feedback.

Key Design Principles. (1) Environment-aware Workflow Adaptation via Strategy Selection. The Meta-Cognitive Controller explicitly assesses both the database environment and the complexity of the input query to determine the appropriate reasoning strategy. For example, for simple queries over well-structured schemas (e.g., single-table filtering), the controller may directly invoke a lightweight generation path using few-shot prompting. In contrast, for complex analytical queries (e.g., multi-join or domain-specific reasoning), the controller

activates a deeper workflow that includes multi-candidate generation, iterative refinement, and optimization loops. This adaptive strategy selection avoids over- or under-utilization of reasoning resources.

(2) Process-level Self-Reflection and Dynamic Control.

Unlike conventional post-hoc validation approaches, our system enables continuous monitoring of intermediate reasoning states during SQL generation. The Meta-Cognitive Controller observes signals such as schema linking ambiguity, candidate disagreement across generated SQLs, and convergence behavior in optimization loops. For instance, if multiple candidate SQLs consistently disagree on join paths, the controller may trigger re-linking of schema entities or restart generation with alternative evidence. Similarly, if repeated optimization iterations fail to reduce execution cost, the controller may terminate the current loop and switch to a different reasoning configuration. This enables closed-loop control over the entire reasoning trajectory.

(3) Execution- and Plan-aware Optimization via Feedback-driven Rewriting. The system treats execution results, query plans, and cost estimations as fundamental feedback signals that continuously shape its optimization behavior. When execution traces reveal inefficiencies such as full table scans, suboptimal join orders, or redundant computation paths, the system invokes an optimization agent that reasons over both the current execution feedback and previously observed optimization experiences to produce improved SQL variants. Crucially, optimization is not performed in isolation for each query. Instead, the system continuously accumulates experience from execution outcomes and rewriting actions, forming an evolving Auto-Evaluation Knowledge Base (AEKB). This knowledge base stores historical rewrite cases, execution feedback, and observed performance patterns, enabling the agent to generalize from past optimization experiences to new queries. In this way, query optimization is transformed from a rule-based or purely local rewriting process into an experience-driven learning loop, where execution feedback serves as the foundation for progressively improving future reasoning and optimization decisions.

III. OPTSQL

IV. EXPERIMENT

V. DISCUSSION

VI. RELATED WORK

Fine-tuned and Trained Text-to-SQL Models. A dominant line of recent work focuses on adapting or training LLMs for Text-to-SQL via supervised fine-tuning, synthetic data construction, or reinforcement learning [33], [20], [34], [35], [36], [37], [38], [39]. For example, XiYAN-SQL [38] uses multi-task fine-tuning and a multi-generator ensemble to produce diverse SQL candidates, while OMNISQL [39] improves model training through large-scale synthesized Text-to-SQL data. Reinforcement-learning-based methods such as REASONING-SQL [40] and ARCTIC-TEXT2SQL-R1 [41]

further train models with SQL-tailored rewards or execution-based feedback to improve reasoning and executability. Despite their strong performance on benchmarks, these methods remain heavily dependent on training data quality and coverage. They often exhibit limited generalization in out-of-domain scenarios and may suffer from reduced robustness when facing complex or real-world database environments [42].

LLM Prompting and Retrieval-augmented Generation.

Another line of work leverages general-purpose LLMs for Text-to-SQL through prompting and retrieval-augmented generation, performing task adaptation at inference time without parameter updates. These methods formulate SQL generation as an in-context reasoning problem, where the model is guided by schema serialization, database content, demonstrations and decomposition instructions. Early benchmark and adaptation studies show that LLMs can become strong Text-to-SQL generators when prompts provide sufficient database context and carefully selected examples [6], [5], [43]. Subsequent prompting methods improve robustness through task decomposition, self-correction, dynamic few-shot selection, and consistency alignment [7], [44], [45]. Retrieval-augmented approaches further strengthen grounding by retrieving relevant demonstrations, linking schema elements at scale, pruning database context, and selecting among multiple candidates [10], [9], [46], [47], [48]. Despite these advances, prompting and retrieval-based methods still depend heavily on manually designed inference procedures and the quality of retrieved evidence. Their reasoning depth is usually determined before generation, which limits their ability to adapt the workflow according to schema scale, intermediate uncertainty, or execution behavior [42], [9].

Agentic Text-to-SQL workflows. Recent work further extends LLM prompting into agentic workflows, where LLMs call tools, inspect database context, execute SQL, observe feedback, and revise intermediate outputs. General agent frameworks such as REACT [17], SELF-REFINE [49], and REFLEXION [50] demonstrate the value of combining reasoning, acting, and iterative self-improvement. Following this paradigm, Text-to-SQL agents organize generation around modules such as schema exploration, candidate generation, execution-based repair, multi-path reasoning, semantic memory, and agentic view construction [8], [46], [51], [13], [14], [15], [11], [12]. These systems are more robust than single-shot prompting because they can query the database, compare alternatives, and use observed failures to revise SQL. Nevertheless, most existing agents still instantiate a largely fixed workflow: similar modules are invoked for most questions, and database feedback is mainly used for post-generation repair, validation, or candidate selection rather than for continuous process control.

Difference from OPTSQL. Existing prompting-based and agentic Text-to-SQL methods mainly focus on improving correctness through better grounding, decomposition, and post-hoc refinement. In contrast, OPTSQL is designed as an efficiency- and process-aware framework that goes beyond static or fixed workflows. It introduces a meta-cognitive con-

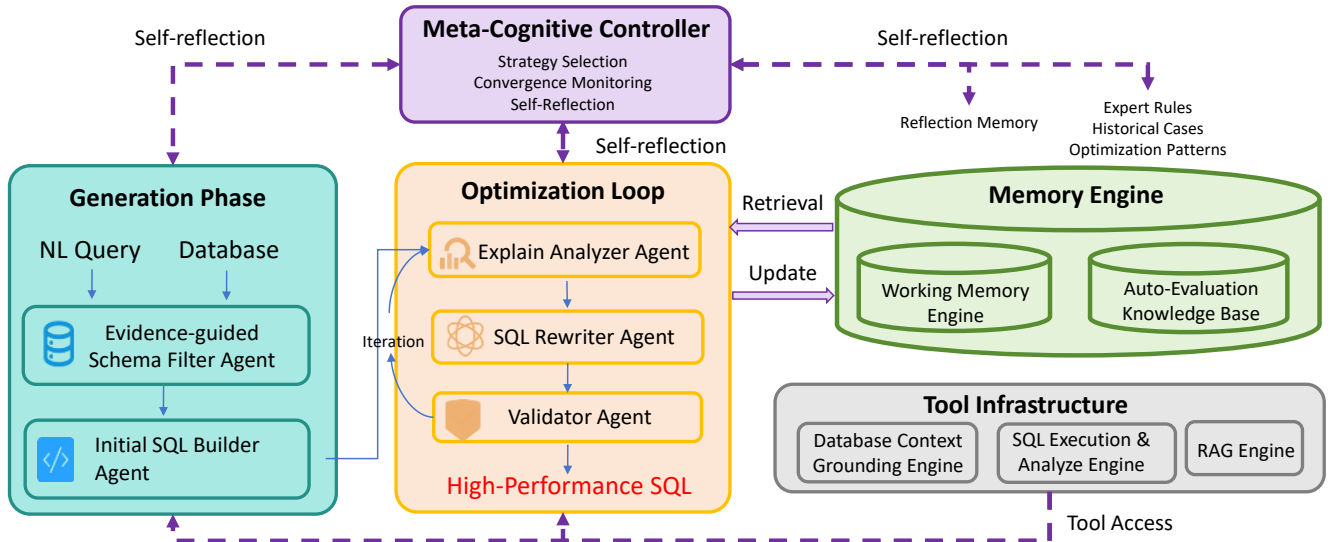


Fig. 2: Overview of OPTSQL

troller that enables (i) environment-aware workflow adaptation, (ii) process-level self-reflection during generation, and (iii) execution- and plan-aware optimization guided by database feedback. As a result, OPTSQL treats Text-to-SQL as a dynamic interaction process where both reasoning strategy and optimization behavior are continuously adjusted according to the database environment and intermediate execution signals, rather than following a predefined pipeline or relying only on post-generation correction.

VII. CONCLUSION

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